COS30082
Applied Machine Learning

Lecture 2

Overview of Machine Learning Algorithms

2.3 Support Vector Machine (SVM)

SVM Topics

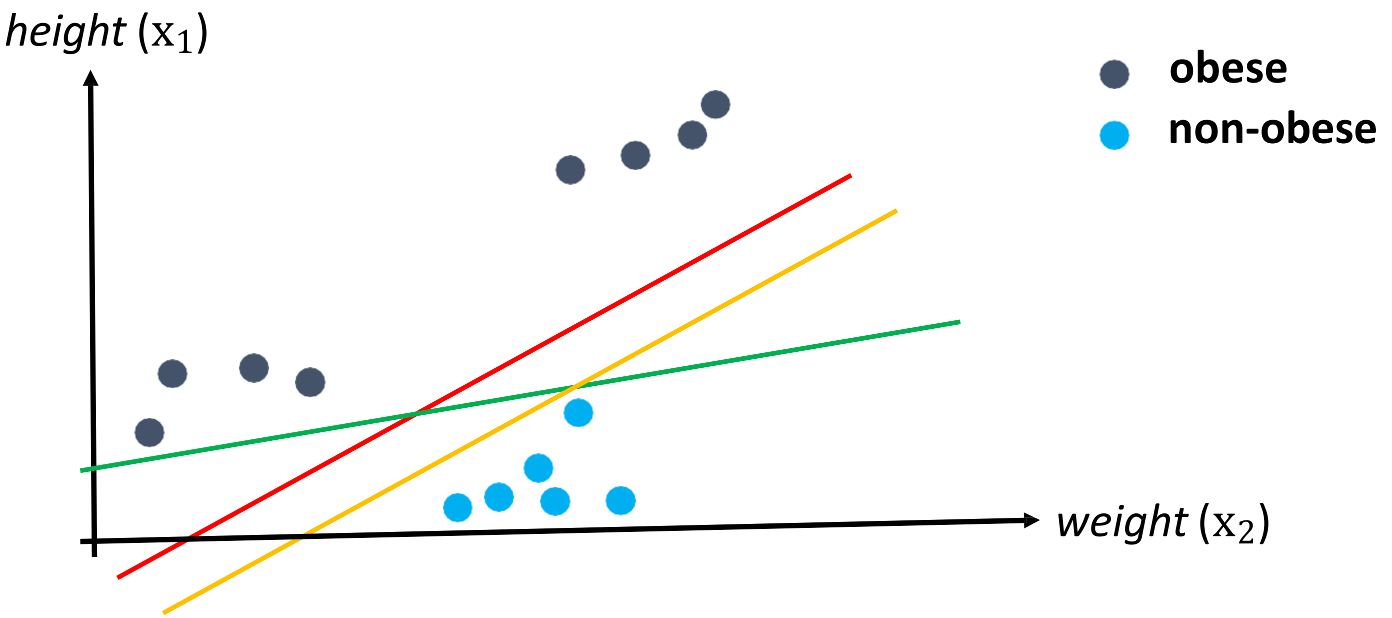
- Understanding of Support Vector Machine (SVM)
- To find out how SVM works
- To learn large-margin optimization techniques
- To explore the differences between SVM and Logistic Regression.
- To solve a machine learning problem using SVM.

What is SVM?

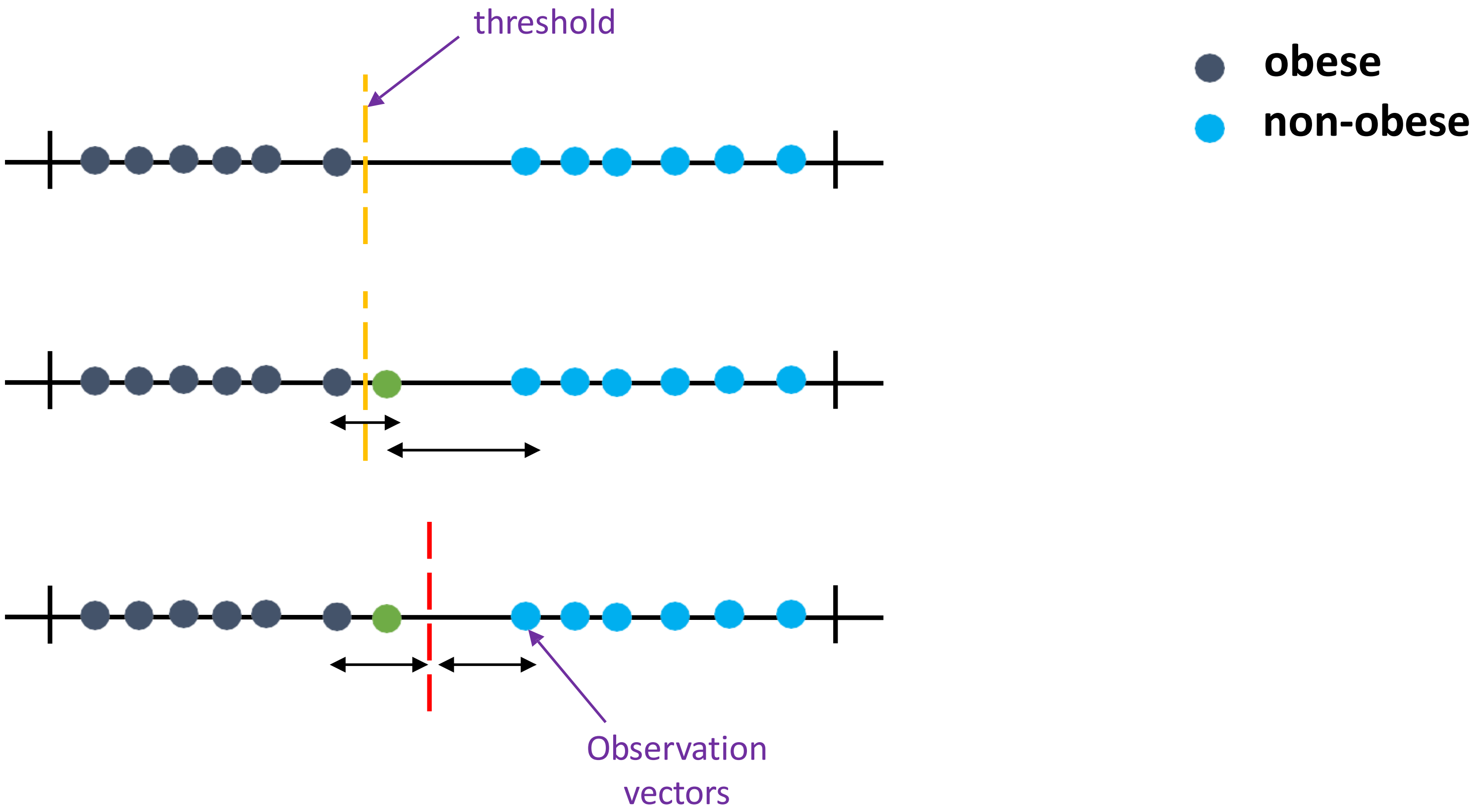
- SVM is a supervised machine learning algorithm which is well-known for classification problem.
- It is also called a **large-margin classifier**

Why large-margin classifier?

- Given *N* number of data points (training set)
- To classify *obese/non-obese* (y) based on the *height* x_1 and *weight* x_2 of the person, which will be the decision boundary chosen by the SVM?



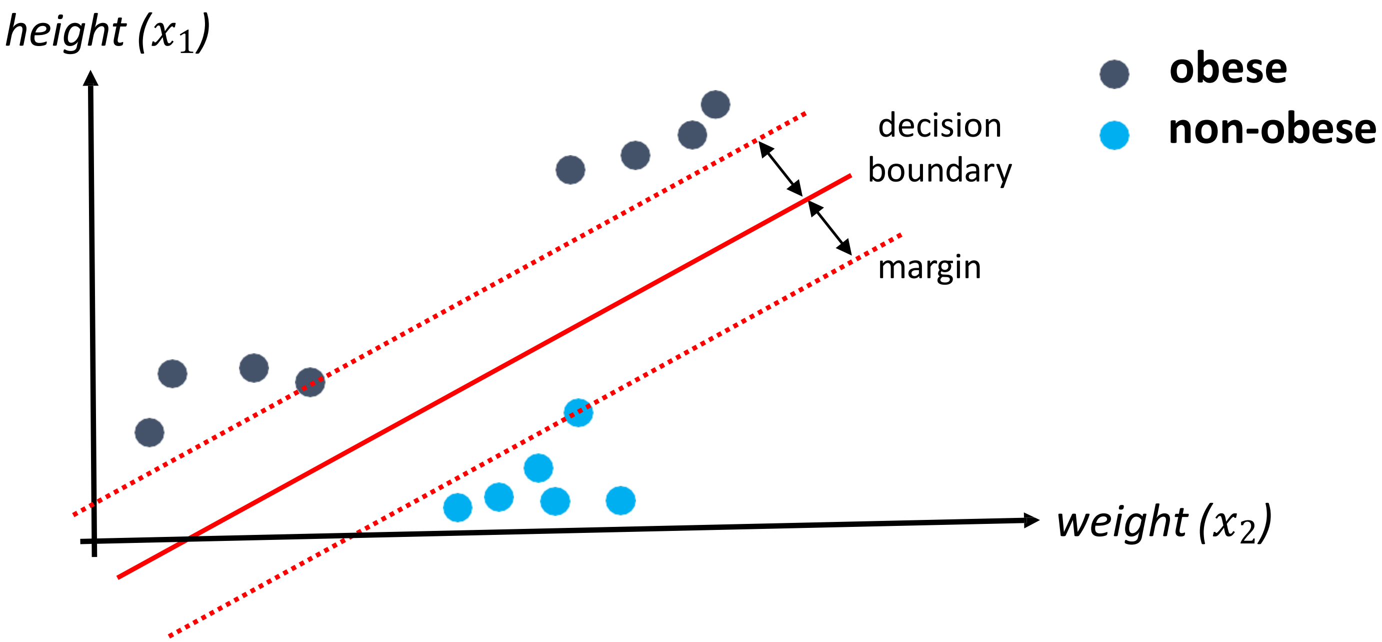
Why large-margin classifier?



- In SVMs, the distance between the observation vectors of the two domains (positive and negative) is equal to the threshold, which together constitute the margin.

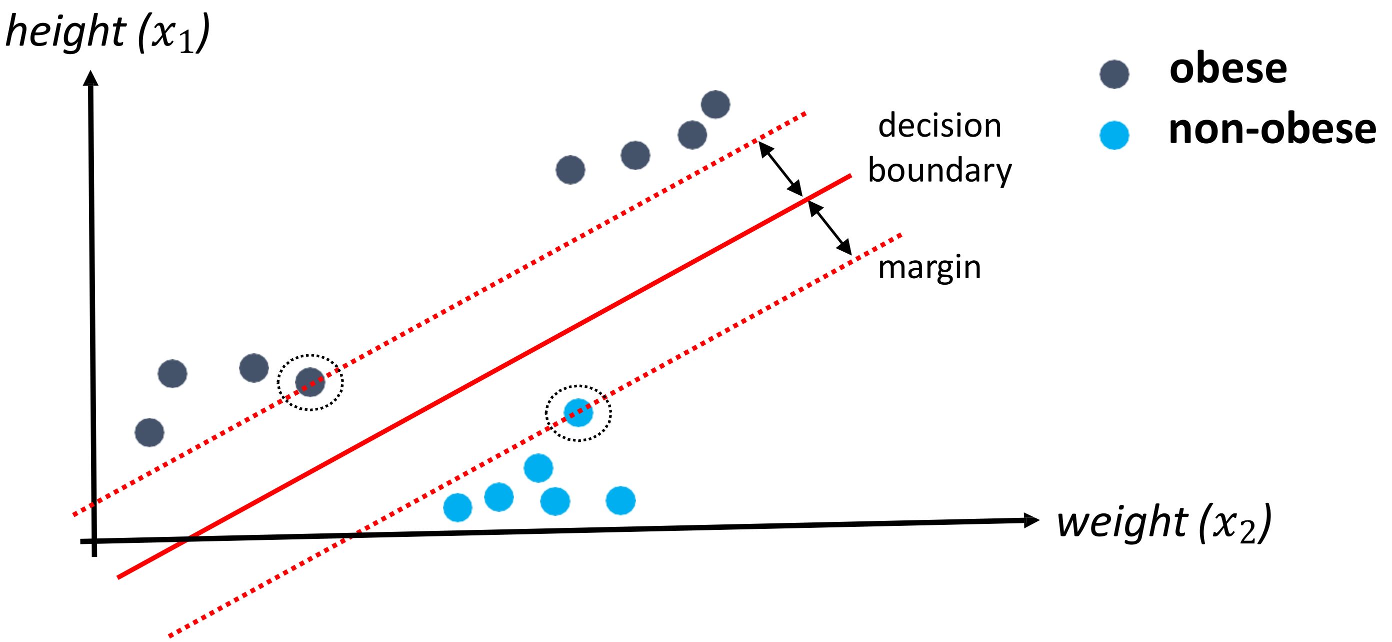
Why large-margin classifier?

- Answer is the red line.
- The objective of the SVM algorithm is to find a decision boundary with the maximum margin, which is the greatest distance from the nearest data point to the decision boundary.



How SVM Works?

- Objective: To find the hyperplane that best separates the classes with the maximum margin.
- Margin: Distance between the hyperplane and the nearest data points from each class (support vectors).



How SVM Works?

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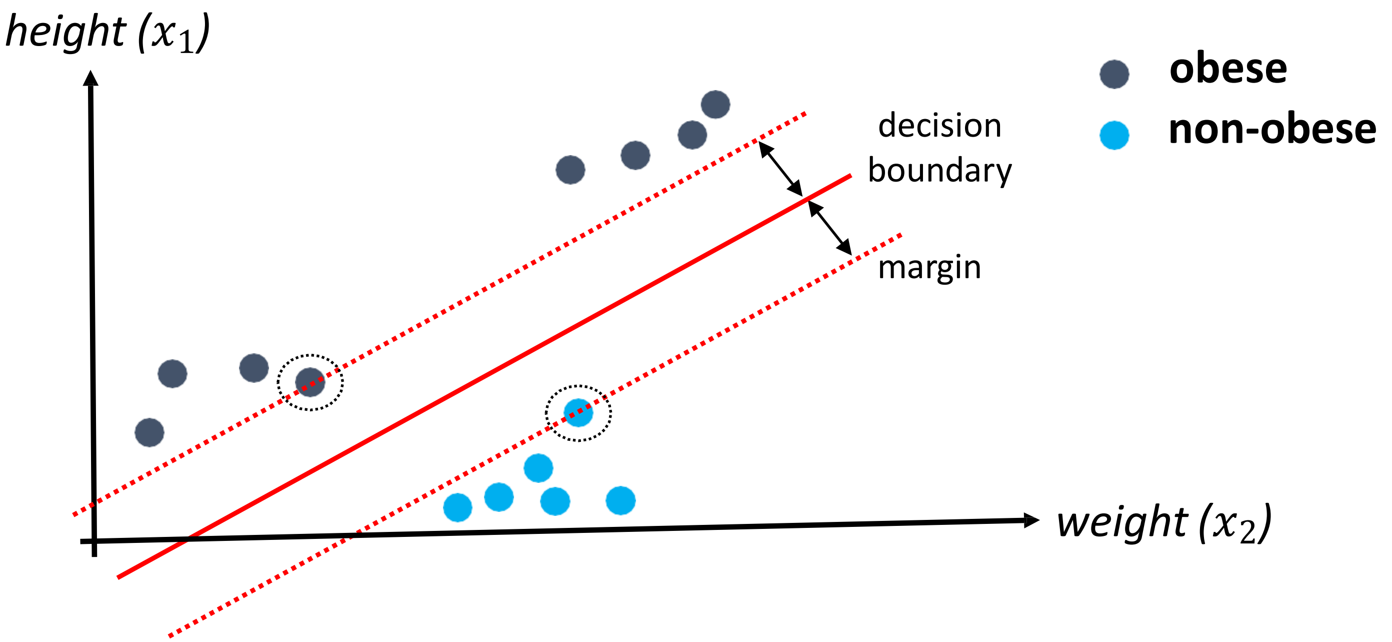
- For linearly separable data, SVM finds a direct hyperplane that separates the classes.
- For non-linearly separable data, SVM uses kernel tricks to transform the data into a higher dimension where a hyperplane can be used for separation.



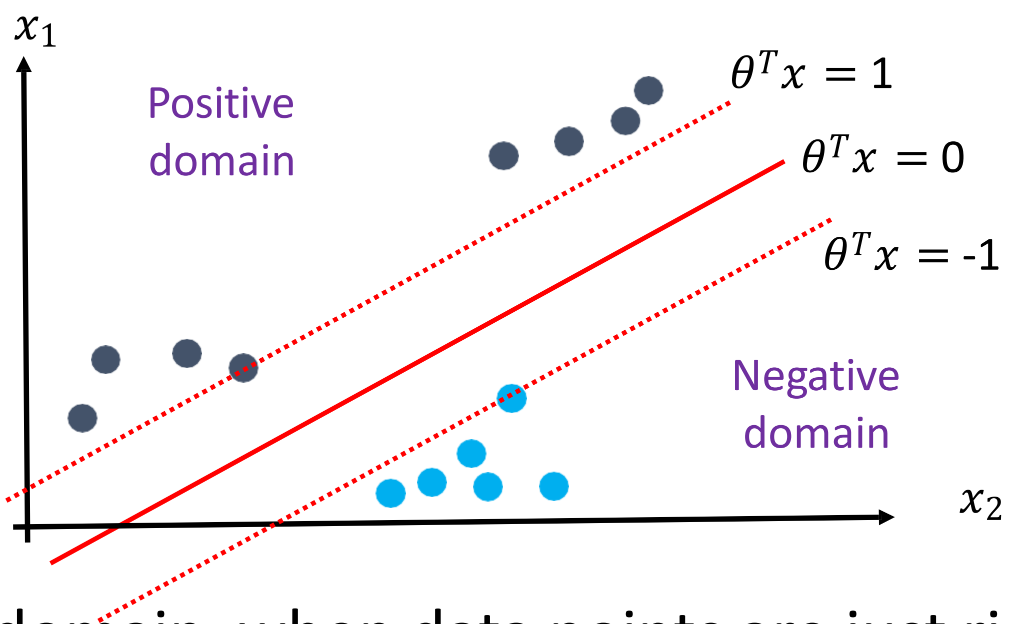
SVM Kernal Trick

- The kernel trick involves transforming data into a higher dimension where it is easier to separate using a linear classifier.
- Common Kernels: Linear, Polynomial, Radial Basis Function (RBF), Sigmoid.
- The choice of kernel affects the decision boundary and the performance of the SVM model.

Math intuition behind SVM

$$if \quad y = 1 \qquad cost_1(\theta^T x) = \max(0, 1 - \theta^T x)$$

$$if \quad y = 0 \qquad cost_0(\theta^T x) = \max(0, 1 + \theta^T x)$$



Assume 3 hyperplanes

- Decision boundary: $\theta^T x = 0$
- Margin: $\theta^T x = \pm 1$

- For the positive domain, when data points are just right on the margin, $\theta^T x$ = 1, when data points are between decision boundary and margin, 0 < $\theta^T x$ < 1.
- For the negative domain, when data points are just right on the margin, $\theta^T x$ = -1, when data points are between decision boundary and margin, -1 < $\theta^T x$ < 0.

Internal

Loss Function

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- The loss function is designed to find a decision boundary (a hyperplane in the feature space) that maximizes the margin between different classes while penalizing points that fall on the wrong side of the margin.
- The two primary components of the SVM loss function are:
  - **Hinge Loss**
  - **Regularization Term**

Hinge Loss

- **Hinge Loss:** The hinge loss is used for "maximum-margin". For an individual sample, the hinge loss is defined as:

$$L_i = \max(0, 1 - y_i(w \cdot x_i - b))$$

- where *w* represents the weights of the model, *xi* is the feature vector of the i-th sample, *b* is the bias term, and *yi* is the actual class label for the i-th sample, which should be -1 or 1. The term *w·xi–b* is the raw output of the classifier. If this output is correct and beyond the margin, the loss is 0. Otherwise, the loss increases linearly with the distance from the margin.

Regularization Term

- **Regularization Term:** To prevent the model from overfitting, a regularization term is added to the loss function. The most common form in SVMs is the $L2$ regularization, which is the squared magnitude of the weight vector w.

Loss Function

- complete SVM loss function with the regularization term is given by:

$$J(w) = \frac{\lambda}{2}||w||^2 + \frac{1}{N}\sum_{i=1}^{N} \max(0, 1 - y_i(w \cdot x_i - b))$$

- where λ is a regularization parameter that controls the trade-off between increasing the margin size and ensuring that the *xi* lies on the correct side of the margin. *N* is the number of training samples.
- A smaller value for λ implies less regularization (allowing more complexity in the model), whereas a larger λ promotes simplicity in the model at the risk of not capturing all the nuances in the data.

Loss Function

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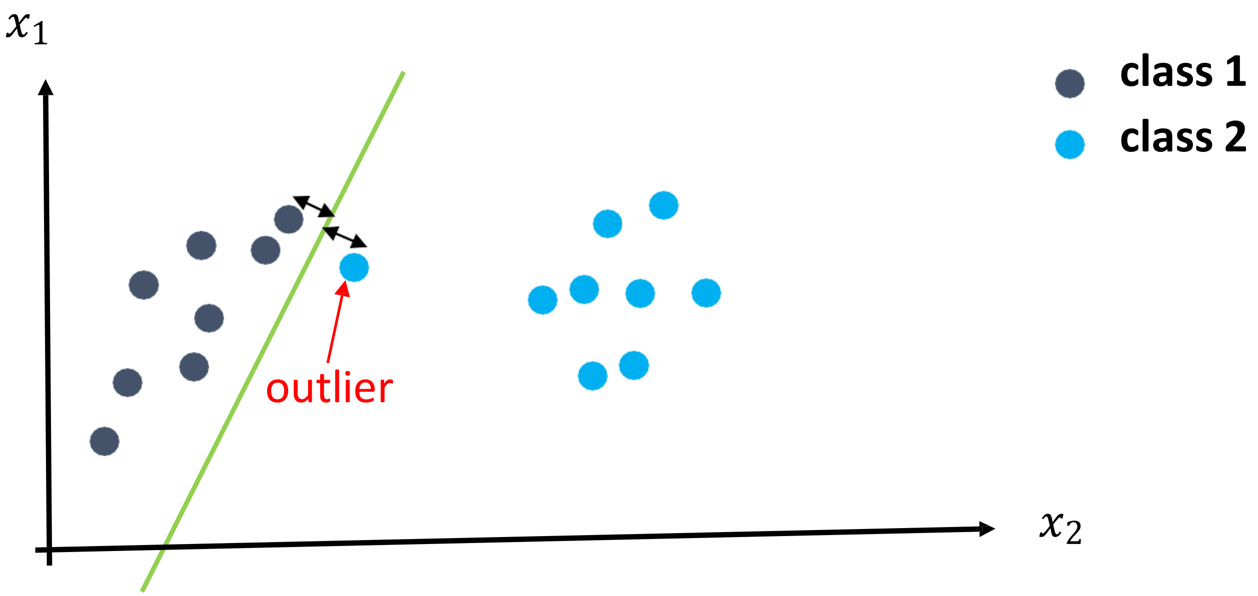
- In In many descriptions of SVMs, particularly in the context of scikit-learn or other practical machine learning libraries, you might encounter the regularization parameter denoted as C rather than λ. The formulation with C looks like this: :
- $$J(w) = C\sum_{i=1}^{N} \max(0, 1 - y_i(w \cdot x_i - b)) + \frac{1}{2}||w||^2$$
- A large value of *C* corresponds to assigning a higher penalty to misclassified points, essentially prioritizing the correctness of classification over the width of the margin. Conversely, a smaller *C* value emphasizes a larger margin and allows more misclassifications.

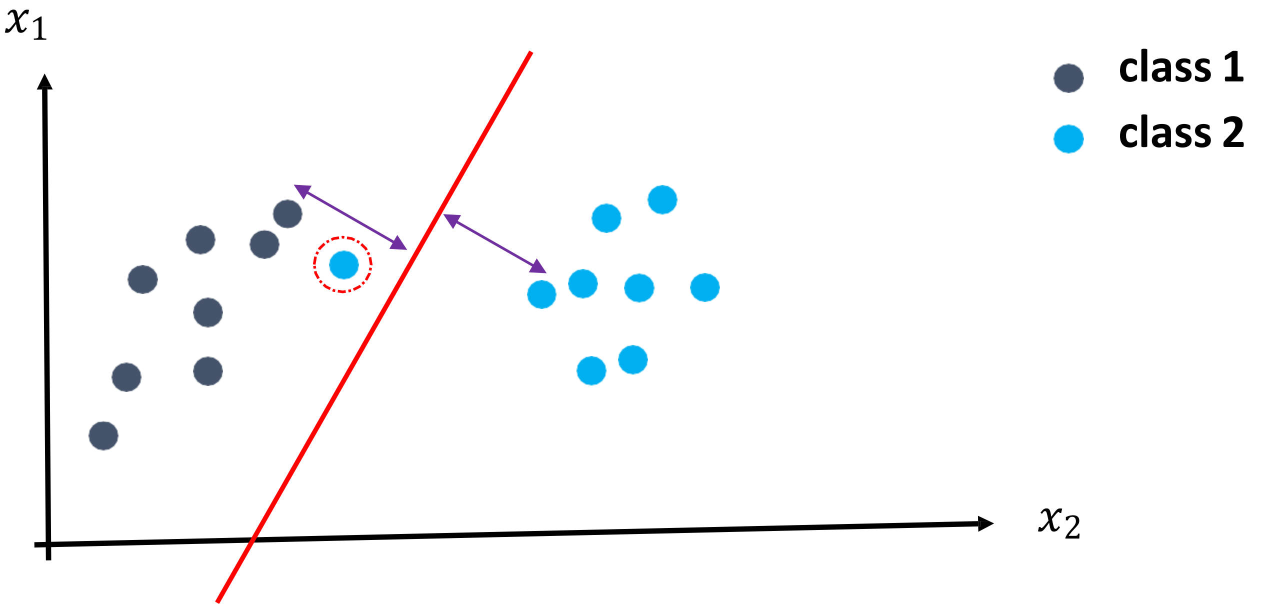
Large C effect

- However, model with large C value is **sensitive to outliers**, similar to no regularization.
- Such a model is subject to overfitting.
  - lower bias, higher variance



Small C effect

- Smaller C allows a bigger margin. It is useful for non-linearly separable dataset with tolerance of data points who are **misclassified** or **have margin violation**.



- However, a model with a too small C value is subject to underfitting
  - Higher bias, lower variance

Non-linear SVM

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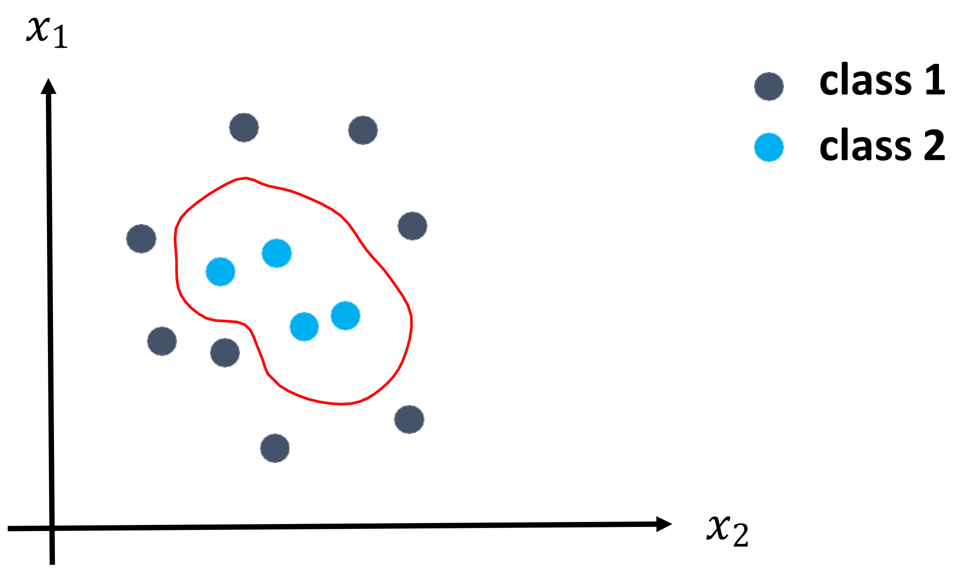
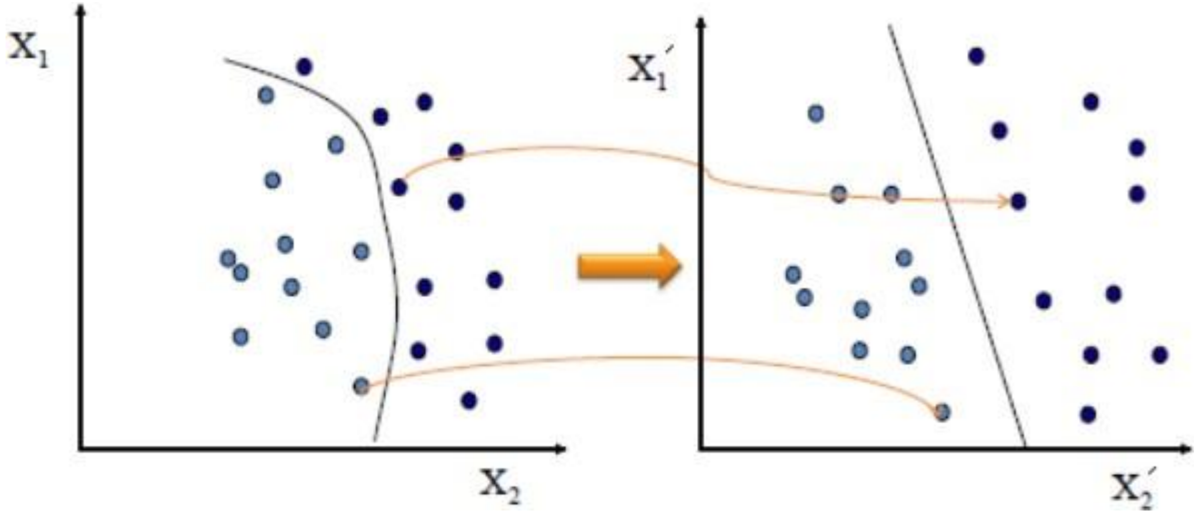
- The simplest way to separate two groups of data is with a straight line (1 dimension), flat plane (2 dimensions) or an N-dimensional hyperplane.
- However, there are situations where a nonlinear region can separate the groups more efficiently.
- Non linear decision boundary is necessary to separate the data points.



What is function f?

- f is a **similarity function** which is called a **kernel function**. It is used to map the data into a different space when a hyperplane (linear) cannot be used to do the separation.
- A non-linear function is learned by a linear learning machine in a high-dimensional feature space while the capacity of the system is controlled by a parameter that does not depend on the dimensionality of the space.
- This is called *kernel trick* which means the kernel function transform the data into a higher dimensional feature space to make it possible to perform the linear separation.



SVM vs logistic regression

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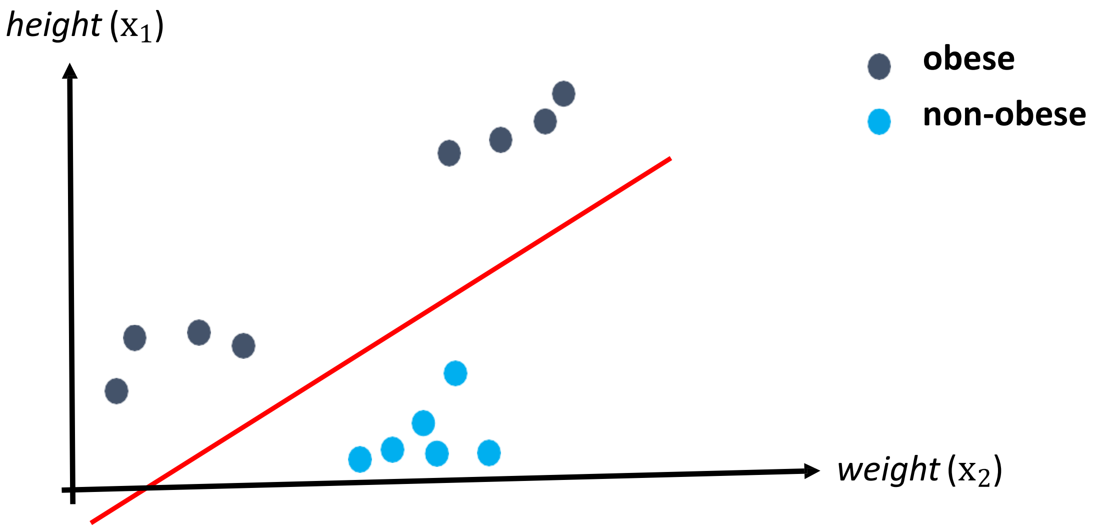
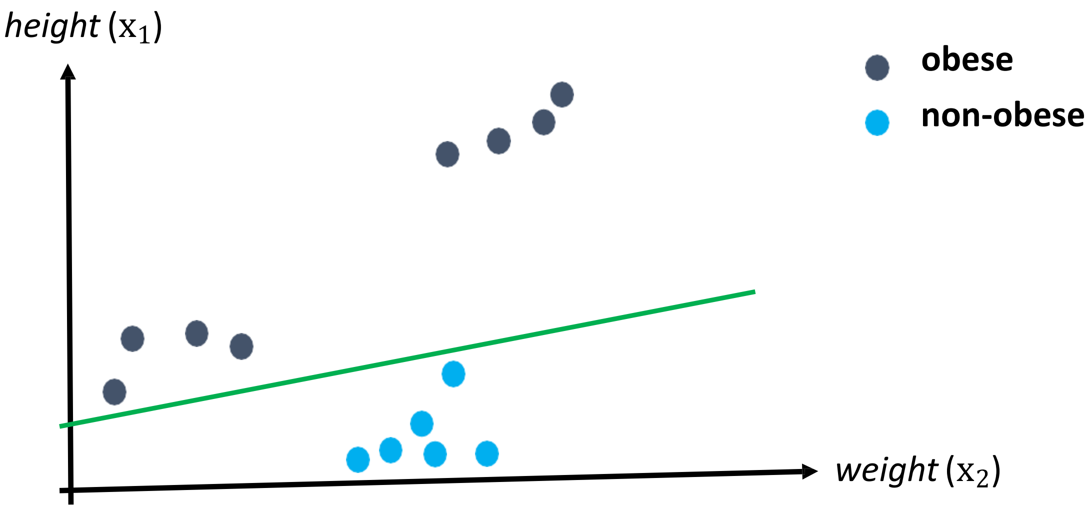
SVM

- It is able to find the maximum margin (distance between the decision boundary and the support vectors) that separates the 2 classes.

Logistic regression (LR)

- It can have different decision boundaries with different parameters θs that are near the optimal solution.





SVM vs logistic regression

- When to use logistic regression and support vector machine?
  - Based on the number of training samples (n) and the number of features (m).

- If m is large and n is small where m >> n: Linear SVM or Logistic Regression might be a choice.
- If m is small and n is intermediate where n > m: SVM with Gaussian kernel might be a choice
- If m is small and n is large where n >> m: Linear SVM or Logistic Regression might be a choice. (preferable add more features)